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NeuroFetal-AI: A Tri-Modal Architecture with Cross-Modal Attention for Cardiotocography Classification

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***Abstract*—Intrapartum fetal compromise—encompassing hypoxia and acidosis during labor—remains a principal cause of perinatal mortality, stillbirth, and long-term neonatal neurological injury worldwide, with disproportionate impact in low-and-middle-income countries where trained obstetric personnel are scarce. Cardiotocography (CTG), the clinical gold standard for intrapartum monitoring, suffers from poor inter-observer agreement (60–70%) and high false-positive rates, motivating the need for reliable, automated decision support. This paper presents NeuroFetal AI v5.0, a clinically deployable, uncertainty-aware decision support system that performs tri-modal fusion of Fetal Heart Rate (FHR) time-series, Uterine Contraction (UC) signals, and 16-dimensional maternal clinical tabular features. The core prediction engine is a Stacking Ensemble of three architecturally diverse base learners: (i) *AttentionFusionResNet*, a six-block residual network augmented with Squeeze-and-Excitation (SE) attention and a Cross-Modal Attention fusion layer; (ii) *1D-InceptionNet*, a multi-scale convolutional network with parallel kernel branches; and (iii) *XGBoost*, a gradient-boosted decision tree trained on 35 hand-crafted tabular and Common Spatial Pattern (CSP) features. A Logistic Regression meta-learner combines out-of-fold predictions via rank-averaged stacking. To address the severe class imbalance (7.25% pathological), we introduce a *TimeGAN*-based augmentation strategy—a Wasserstein GAN with gradient penalty (WGAN-GP) trained exclusively on pathological FHR+UC traces—generating 1,410 physiologically coherent synthetic minority-class recordings that preserve temporal dynamics such as late decelerations and contraction timing. Model calibration is performed via Platt Scaling, and epistemic uncertainty is quantified through Monte Carlo (MC) Dropout combined with information-theoretic metrics (Predictive Entropy and Mutual Information). Evaluated on the public CTU-UHB Intrapartum Cardiotocography Database (552 recordings, PhysioNet) under stratified 5-fold cross-validation, NeuroFetal AI v5.0 achieves AUC 0.8639, 96.34% Accuracy, 95.22% F1-Score, Brier Score 0.046, and ECE 0.0543, surpassing the state-of-the-art baseline of Mendis et al. (AUC 0.84) which relied on an unpublished private dataset of over 9,800 recordings. The system further provides Grad-CAM signal-level explainability and is packaged as a 1.9 MB TFLite Int8 model for offline inference on commodity mobile hardware, enabling deployment in resource-constrained clinical settings.**

Index Terms—Cardiotocography, Fetal Heart Rate, Fetal Compromise, Deep Learning, Stacking Ensemble, Tri-Modal Fusion, TimeGAN, Uncertainty Quantification, Explainable AI, Edge Deployment, Clinical Decision Support.

I. Problem Statement

A. Clinical Context and Motivation

Approximately **2.6 million stillbirths** occur globally each year, with the vast majority concentrated in low-resource settings where access to trained obstetric specialists is critically limited [1]. A significant proportion of these adverse outcomes are attributable to undetected or late-detected intrapartum fetal compromise—a condition characterized by progressive fetal hypoxia and metabolic acidosis during labor, resulting from insufficient uteroplacental oxygen delivery. Catching fetal compromise early and accurately is essential because it gives doctors time to intervene—whether through an emergency cesarean section or instrumental delivery. Such timely actions are often the only way to prevent fetal death or permanent neonatal brain damage.

B. Cardiotocography: The Clinical Standard and Its Limitations

Clinicians virtually everywhere rely on **Cardiotocography (CTG)** as the standard approach for keeping track of fetal health during labor. It simultaneously records two signals: the Fetal Heart Rate (FHR), measured by an ultrasound Doppler transducer on the maternal abdomen, and Uterine Contractions (UC), measured via tocodynamometry. To interpret these readings, doctors look for specific patterns in the FHR, such as baseline rate, short-term and long-term variability (STV

and LTV), accelerations, and decelerations. They then analyze these patterns against the timeline of uterine contractions. Guidelines from the International Federation of Gynecology and Obstetrics (FIGO) classify CTG traces as Normal, Suspicious, or Pathological [2].

However, visual CTG analysis is hindered by several limitations:

1) **Poor inter-observer agreement**: Multiple studies demonstrate only 60–70% agreement between experienced obstetric clinicians interpreting the same CTG trace, reflecting the inherent subjectivity of visual pattern recognition [3].
2) **Alert fatigue and high false-positive rates**: Existing automated CTG analysis systems generate excessive false alarms, leading clinicians to habitually override or dismiss system warnings—a phenomenon that undermines system utility and patient safety.
3) **Single-signal dependency**: Current commercial and research automated systems predominantly analyze the FHR signal in isolation, disregarding the clinically essential temporal relationship between FHR patterns and uterine contraction timing. Late decelerations—FHR drops occurring after the contraction peak—are the cardinal marker of uteroplacental insufficiency and can only be identified through joint FHR-UC analysis.

C. Limitations of Existing Automated Approaches

Prior deep learning approaches to automated CTG analysis exhibit several critical deficiencies that limit their clinical applicability:

- **Unimodal signal analysis**: Most published deep learning systems, including early 1D-CNN and ResNet-based approaches [8], [9], process FHR in isolation, omitting the UC signal and all clinical metadata.
- **Absence of uncertainty quantification**: Existing models produce deterministic point predictions without confidence estimates. In a clinical context, a prediction delivered without a reliability measure is inherently dangerous, as it provides no mechanism for the clinician to identify ambiguous or out-of-distribution cases requiring human review.
- **Reliance on large private datasets**: State-of-the-art results, such as those reported by Mendis et al. [4] (AUC 0.84), were achieved using a private institutional dataset of 9,887 recordings—far exceeding the size of any publicly available CTG database. This places their results beyond reproducibility and limits comparative benchmarking within the research community.
- **Deployment inaccessibility**: Published models are typically evaluated in laboratory settings and are not optimized for deployment on resource-constrained hardware. In low-income clinical settings, inference must be possible on commodity hardware without persistent internet connectivity.

D. Research Problem Definition

Given the foregoing limitations, this work addresses the following research problem:

How can a multi-modal deep learning system integrate FHR time-series, UC signals, and maternal clinical tabular features to accurately detect intrapartum fetal compromise from a small public dataset, while providing clinically meaningful uncertainty estimates, transparent decision explanations, and remaining deployable on edge hardware—thereby enabling practical use in resource-limited obstetric settings?

The specific research objectives are:

1) To design a tri-modal fusion architecture that exploits the complementary information in FHR signals, UC signals, and structured clinical tabular data for fetal compromise classification.
2) To overcome documented class imbalance (7.25% pathological) and limited dataset size (552 recordings) through a physiologically realistic time-series generative augmentation strategy (TimeGAN).
3) To build a clinically trustworthy system via probabilistic uncertainty quantification (MC Dropout, Platt Scaling) and signal-level explainability (Grad-CAM).
4) To achieve state-of-the-art performance on the publicly available CTU-UHB benchmark, outperforming results obtained with private datasets, thereby establishing a reproducible baseline for the community.
5) To compress and quantize the trained model to under 2 MB for real-time offline inference on mobile devices.

II. Literature Survey

Historically, researchers trying to automate CTG analysis have gone through three major phases. Initially, they focused on manually designing features, then moved on to classical machine learning techniques, and most recently, they have embraced deep learning. In this section, we take a look at some of the most impactful studies to show exactly where NeuroFetal AI fits in.

A. Classical Feature-Engineering and Machine Learning Approaches

Early automated CTG analysis systems relied on expert-defined morphological features derived from FHR traces. Surveys such as those by Georgoulas et al. [10] and Spilka et al. used predefined feature sets such as baseline FHR, STV, LTV, accelerations, and deceleration counts. These features were then fed to classical classifiers:

Support Vector Machines (SVM) formed the backbone of early competitive systems, achieving AUC scores in the range 0.72–0.79 on the CTU-UHB database [7]. Fergus et al. [11] and Czabanski et al. employed SVM with radial basis function kernels and feature selection to identify pathological traces, demonstrating that clinical features such as baseline rate and STV were the most discriminative.

Random Forests and Ensemble Methods: Gradient-boosting and random forest classifiers applied to hand-crafted FHR features demonstrated improvements over SVM, reaching

AUC values near 0.83 [12], and established that feature ensembling could compensate for the limitations of individual engineered metrics. The real problem with these classical methods, however, was their limited scope. Because they mostly looked at aggregated statistics, they completely missed the subtle, moment-by-moment temporal structure of the FHR sequence. Even worse, they ignored the critical timing between FHR changes and uterine contractions.

B. Deep Learning Approaches: Single-Modal Signal Analysis

Convolutional neural networks (CNNs) have facilitated the extraction of temporal patterns directly from raw FHR sequences, reducing reliance on hand-crafted features.

1D Convolutional Neural Networks: Zhao et al. [9] proposed a 1D-CNN architecture processing raw FHR sequences, demonstrating superior performance over classical baselines on the CTU-UHB dataset. However, evaluation was limited to the FHR channel exclusively, ignoring UC. Building on this, **Petrozziello et al. (2019)** [27] introduced an input-length invariant deep learning approach for FHR, allowing rapid detection of fetal compromise without requiring fixed-length windows, improving potential clinical utility.

Segment selection duration is an important factor in FHR analysis. Recent studies, such as the cross-database evaluation by **Lopes et al. (2025)** [24], have demonstrated that utilizing extended continuous FHR segments (e.g., $\geq$ 30 minutes immediately preceding delivery) and explicitly preserving signal gaps yields stronger generalization performance across diverse clinical datasets compared to aggressively gap-interpolated 10-minute snippets.

Residual Networks (ResNet): Spilka et al. [8] and subsequent work demonstrated that residual connections substantially improved gradient flow during training of deep 1D temporal networks. Mendis et al. [4] employed a three-block 1D-ResNet as the signal-analysis branch of their Fusion ResNet, establishing AUC 0.80 for the FHR-only configuration—a result they subsequently improved to AUC 0.84 by augmenting with tabular clinical features.

Multi-Scale Convolutional Architectures: Inspired by *InceptionNet* [13], several works applied parallel convolutions with varying kernel sizes (capturing patterns at different temporal scales) to FHR classification. Furthermore, attention-based models such as the Multi-Scale CNN (MCNN) [9] achieved AUC 0.81 on CTU-UHB, representing the prior deep learning state-of-the-art before multimodal fusion.

Recurrent Architectures: Long Short-Term Memory (LSTM) networks and gated recurrent units (GRU) have been explored for FHR sequence modeling [14], leveraging their capacity for long-range temporal dependency modeling. However, their sequential computation renders them significantly slower than convolutional approaches for deployment, and they did not consistently outperform well-regularized ResNets on the CTU-UHB benchmark.

Emerging Architectures: Recent advancements have seen the application of **Foundation Models** to CTG. For instance, a 2024 study [28] proposed a foundation model approach pre-trained on massive unlabeled FHR corpora and fine-tuned on CTU-UHB for fetal stress prediction during labor, demonstrating strong zero-shot and few-shot capabilities. Additionally, research expanding beyond just the second stage of labor has shown that deep learning classifiers can predict fetal health across *both* stages of labor [30], though these models often struggle generalizing across diverse hospital protocols.

C. Multimodal and Fusion-Based Approaches

Recognition that FHR alone is insufficient motivated the development of multimodal systems.

Fusion ResNet (Mendis et al., 2023) [4]: This work, which serves as the direct baseline for NeuroFetal AI, presented the most relevant prior architecture. It proposed a late-fusion design combining a three-block 1D-ResNet (processing the last 60 minutes of FHR at 1 Hz) with a two-layer Dense Network processing five tabular clinical features (Parity, Maternal Age, Gestation, Baseline Intercept β_0, and Median Absolute Deviation of detrended FHR). Fusion was performed via element-wise multiplication, which the authors demonstrated outperformed concatenation and addition as fusion operators. The model achieved **AUC 0.84** on the public CTU-UHB dataset and was validated externally on their private MHW-pH dataset (9,887 recordings). Critically, the uterine contraction channel was discarded due to signal quality constraints, representing a data modality that NeuroFetal AI explicitly incorporates. The authors provided SHAP-based tabular feature attribution and Grad-CAM signal localization, establishing dual-XAI as the interpretability standard for this domain.

Christensen et al. and related work explored clinical risk scoring combined with CTG signal features, confirming that maternal clinical covariates (Parity, Age, Gestation) add discriminative value beyond signal morphology alone.

Sadeghi et al. (2024) [25] advanced the multimodal paradigm by developing a deep learning algorithm trained exclusively on the CTU-UHB database to detect specific morphological FIGO events (e.g., variable decelerations) rather than just binary fetal outcomes. This approach indicates a shift toward event-specific detection rather than broad risk scoring.

D. Time-Series Data Augmentation for Medical Imbalance

Class imbalance is an endemic challenge in clinical CTG datasets, where pathological cases comprise fewer than 10% of recordings.

SMOTE [16] applied to flattened FHR feature vectors was a standard baseline augmentation, though it operates in feature space and destroys temporal structure—generating synthetic sequences that may not correspond to physiologically plausible FHR patterns.

TimeGAN (Yoon et al., 2019) [15] introduced a framework for realistic time-series synthesis using a GAN architecture with a supervised loss component preserving stepwise temporal dynamics. Application of TimeGAN to pathological CTG synthesis has not been widely explored in prior work, which

we investigate in this study. NeuroFetal AI employs a WGAN-GP variant of TimeGAN that generates physiologically coherent FHR+UC traces with preserved deceleration timing relative to contractions.

E. Uncertainty Quantification in Clinical AI

MC Dropout (Gal & Ghahramani, 2016) [17]: Demonstrated that Dropout applied at inference time provides a practical Bayesian approximation, yielding predictive uncertainty through variance across stochastic forward passes. This has been applied in medical imaging [18] but remains underexplored in time-series CTG classification.

Calibration: Platt Scaling [19] and isotonic regression are established post-hoc calibration techniques that map model output scores to well-calibrated probabilities. Their application to ensemble CTG classifiers has not been previously reported in the literature.

F. Explainable AI in Fetal Monitoring

Grad-CAM [20] has been adapted to 1D convolutional networks for temporal signal attribution, identifying which time segments most influenced the classification decision. Mendis et al. [4] applied Grad-CAM to their ResNet branch to validate that the model correctly attended to late deceleration patterns. **SHAP** (SHapley Additive exPlanations) [21] provides model-agnostic tabular feature attribution and was used in [4] to explain the contribution of clinical risk factors such as nulliparity and maternal age.

Recent interpretable architectures, such as **DeepCTG 1.0** [29], were specifically designed to provide clinicians with transparent feature attributions for fetal hypoxia detection, reinforcing that high AUC alone is insufficient for clinical adoption; models must provide physiologically plausible rationales. Similarly, researchers have begun hybridizing deep learning with classical signal processing, such as combining Instantaneous Frequency analysis with **Common Spatial Patterns (CSP)** [31] for FHR classification, a method we apply in this study for FHR-UC cross-modal extraction.

G. Edge Deployment of Medical AI

TensorFlow Lite (TFLite) quantization enables compression of deep learning models to megabyte scale with minimal accuracy loss, suitable for deployment on commodity mobile hardware. Its application to deployed obstetric AI systems remains limited in published literature, with most systems assuming continuous hospital network connectivity and GPU hardware.

Addressing this constraint, recent literature has explored the integration of deep learning into mobile clinical interfaces. For example, the **AI-based Mobile Partograph (2025)** [26] explicitly targets resource-limited delivery rooms by embedding a predictive CTG framework (validated on CTU-UHB) directly into a digital labor chart interface. However, such mobile systems currently lack predictive uncertainty quantification, a critical requirement for clinical trust.

H. Research Gap and Positioning of NeuroFetal AI

Table I summarizes the key limitations of prior work and the methods used in this study. Three primary gaps are addressed:

1) *UC signal omission*: All prior deep learning systems discarded the UC channel; NeuroFetal AI incorporates it via Cross-Modal Attention to leverage the FHR–contraction temporal relationship.
2) *No uncertainty quantification*: No prior clinical CTG deep learning system provided calibrated uncertainty estimates; NeuroFetal AI delivers MC Dropout variance, Predictive Entropy, and Mutual Information alongside every prediction.
3) *Public-only reproducibility*: Mendis et al.'s leading result (AUC 0.84) required a private 9,887-case dataset. NeuroFetal AI surpasses this on the public CTU-UHB dataset alone (AUC 0.8639), establishing a reproducible benchmark.

TABLE I
COMPARATIVE ANALYSIS OF CTG DEEP LEARNING SYSTEMS

System	AUC	UC	Uncert.	XAI	Edge
Spilka et al. [7] (SVM)	0.76	✗	✗	✗	✗
Zhao et al. [9] (MCNN)	0.81	✗	✗	✗	✗
Mendis et al. [4]	0.84	✗	✗	✓	✗
Sadeghi et al. [25]	N/A*	✓	✗	✓	✗
NeuroFetal AI v5.0 (Ours)	**0.8639**	✓	✓	✓	✓

*(*Note: [25] detects specific events rather than global AUC.)*

(Note: ✓ = present, ✗ = absent. The symbols ✓ and ✗ require the `pifont` *package: add* `\usepackage{pifont}` *and define* `\cmark`/`\xmark` *accordingly, or replace with ✓ and ×.)*

III. DATASET

A. Source: CTU-UHB Intrapartum Cardiotocography Database

All experiments in this work were conducted exclusively on the **CTU-UHB Intrapartum Cardiotocography Database** [5], a publicly available, fully de-identified research dataset hosted on **PhysioNet** [6]. The database was jointly created by the Czech Technical University (CTU) and the University Hospital Brno (UHB), and is the largest publicly available labeled intrapartum CTG dataset with objective biochemical outcome labels.

1) Dataset Characteristics:

- **Total Recordings**: 552 intrapartum CTG recordings from singleton pregnancies with a cephalic presentation and a gestational age of at least 36 weeks.
- **Signal Channels**: Two channels per recording—Channel 1 (FHR, beats per minute, bpm) and Channel 2 (UC, arbitrary tocodynamometric units).

- **Native Sampling Rate**: 4 Hz (i.e., 4 samples per second, corresponding to 240 samples per minute).
- **Recording Duration**: Variable; clinically the last 60 minutes of the intrapartum period are most diagnostically relevant and are exclusively used in this study.
- **Ground Truth Label**: The binary classification target is derived from umbilical cord arterial blood pH measured immediately post-delivery:

$$y = \begin{cases} 1 & \text{(Compromised)}, \quad \text{pH} < 7.05 \\ 0 & \text{(Normal)}, \quad \text{pH} \geq 7.05 \end{cases} \tag{1}$$

  A pH of 7.05 is the established clinical threshold for significant fetal metabolic acidemia [2].
- **Class Distribution**: 40 recordings (7.25%) are labeled Compromised; 512 recordings (92.75%) are labeled Normal. This constitutes a heavily imbalanced binary classification problem.
- **Clinical Metadata**: Each recording is accompanied by structured clinical metadata including: maternal age (years), gravidity, parity (nulliparous or multiparous), gestational age (weeks), delivery type, and Apgar scores at 1 and 5 minutes.
- **Dataset Availability**: Freely accessible at https://physionet.org/content/ctu-uhb-ctgdb/1.0.0/.
- **Ethics**: The CTU-UHB dataset was collected under appropriate ethical approval at the originating institution and is made available under PhysioNet's data use agreement. No Protected Health Information (PHI) is present. No additional institutional review was required for secondary analysis.

B. Preprocessing Pipeline

Raw CTG recordings in PhysioNet WFDB format (`.dat/.hea`) were processed through a standardized, reproducible pipeline implemented in `data_ingestion.py`:

1) Step 1: Signal Extraction: FHR and UC signals were read using the `wfdb` Python library. The final 60 minutes (14,400 samples at 4 Hz) of each recording were cropped to focus processing on the clinically most informative intrapartum segment.

2) Step 2: FHR Artifact Correction: Signal dropout events (consecutive zero-valued samples representing sensor disconnection) were handled as follows: gaps of fewer than 15 seconds (i.e., < 60 samples at 4 Hz, < 15 samples at 1 Hz) were linearly interpolated. Gaps of 15 seconds or longer were preserved as zero-padded segments, as interpolation over extended dropouts risks introducing physiologically implausible artifacts.

3) Step 3: UC Signal Cleaning: Tocodynamometer signals are notoriously susceptible to maternal movement artefacts and baseline drift. A dedicated UC cleaning pipeline (`uc_cleaning.py`) applied median baseline subtraction and amplitude normalization to improve contraction detection reliability.

4) Step 4: Downsampling: Both FHR and UC channels were resampled from their native 4 Hz to **1 Hz** via linear interpolation, consistent with the approach in [4]. This reduces sequence length by 4× (60 min × 60 s × 1 Hz = 3,600 samples per recording), lowering computational cost while preserving diagnostically relevant FHR morphological features.

5) Step 5: MinMax Normalization: FHR and UC signals were independently scaled to the range $[0, 1]$:

$$x_{\text{norm}} = \frac{x - x_{\min}}{x_{\max} - x_{\min}} \tag{2}$$

Normalization was computed per-recording to remove inter-patient amplitude variation and was applied prior to windowing.

6) Step 6: Sliding Window Segmentation: Given the 60-minute (3,600 samples at 1 Hz) signal per recording, a sliding window strategy was employed to expand the effective dataset size:

- **Window Length**: 20 minutes (1,200 samples at 1 Hz).
- **Stride**: 10 minutes (600 samples), yielding a 50% overlap between consecutive windows.
- **Label Inheritance**: Each window inherits the pH-derived binary label of its parent recording.
- **Result**: ∼5 windows per recording, expanding 552 recordings to approximately **2,760 training samples**—a 5× effective expansion of the dataset.

Window shapes are: FHR window $\in \mathbb{R}^{1200\times1}$, UC window $\in \mathbb{R}^{1200\times1}$.

7) Step 7: Tabular Feature Extraction: For each 20-minute window, 16 structured clinical features were extracted across two categories:

Demographic Features (3):

1) *Maternal Age* (years, continuous) — Risk factor for uteroplacental insufficiency.
2) *Parity* (binary: 0 = Nulliparous, 1 = Multiparous) — Nulliparity is strongly associated with prolonged labor.
3) *Gestational Age* (weeks, continuous) — Determinant of fetal maturity and expected FHR baseline range.

Signal-Derived Features (13): Per-window statistics computed from the FHR signal:

1) *Baseline FHR* (bpm) — Estimated mean FHR, excluding accelerations and decelerations.
2) *Short-Term Variability (STV)* — RMS of beat-to-beat differences; a critical marker of fetal neurological integrity.
3) *Long-Term Variability (LTV)* — Amplitude range of oscillatory cycles; reduced LTV is associated with fetal compromise.
4) *Acceleration Count* — Number of FHR rises ≥ 15 bpm above baseline for ≥ 15 seconds.
5) *Deceleration Count* — Number of FHR drops below baseline.
6) *Late Deceleration Flag* — Binary indicator of decelerations temporally lagging contraction peaks (key marker of uteroplacental insufficiency).

7) *Variable Deceleration Count* — Abrupt FHR drops, typically cord compression indicators.
8) *Approximate Entropy (ApEn)* — Non-linear regularity measure of FHR signal complexity.
9) *Sample Entropy (SampEn)* — Conditional probability of pattern continuation; complement to ApEn.
10) *UC Frequency* — Number of detectable contraction events per window.
11) *UC Amplitude* — Average peak-to-trough amplitude of contraction waveforms.
12) *FHR–UC Lag* — Cross-correlation lag at peak FHR-UC correlation, distinguishing early, variable, and late deceleration patterns.
13) *Signal Quality Index* — Percentage of non-artefact samples in the FHR window.

Missing values in tabular features (arising from recordings with insufficient signal quality for feature computation) were imputed using column-wise means computed from the training fold.

8) Step 8: Common Spatial Pattern (CSP) Feature Extraction: A complementary feature representation was derived using **Common Spatial Patterns (CSP)**, a technique originally developed for electroencephalographic (EEG) brain-computer interface (BCI) analysis, here applied as a novel modality for fetal monitoring. CSP identifies spatial filters that maximize the ratio of the variance of one class's signal to the other, applied to the two-channel (FHR, UC) input:

$$W = \text{CSP}(X_{\text{FHR}}, X_{\text{UC}}) \in \mathbb{R}^d \tag{3}$$

where W comprises variance projections of FHR+UC interactions. This produces a **19-dimensional CSP feature vector** per window, providing a compact representation of the discriminative variance structure in the FHR–UC interaction that is not captured by either signal alone. CSP filters were fitted exclusively on the training fold within each cross-validation iteration to prevent data leakage.

C. Processed Dataset Summary

TABLE II
PROCESSED DATASET SPECIFICATIONS FOR NEUROFETAL AI TRAINING

Array File	Shape	Description
`X_fhr.npy`	$(N, 1200, 1)$	FHR windows at 1 Hz, 20 min
`X_uc.npy`	$(N, 1200, 1)$	UC windows at 1 Hz, 20 min
`X_tabular.npy`	$(N, 16)$	3 demographic + 13 signal-derived
`X_csp.npy`	$(N, 19)$	CSP variance features
`y.npy`	$(N,)$	Binary labels (0=Normal, 1=Comp.)
Approx. N (before augmentation)		∼2,760
Pathological windows (before aug.)		∼200 (7.25%)

D. TimeGAN Data Augmentation Strategy

The extreme class imbalance (7.25% pathological) necessitated a robust augmentation strategy. NeuroFetal AI v4.0 replaced the previously employed SMOTE with **TimeGAN**, a time-series generative adversarial network [15] implemented as a WGAN-GP architecture:

- **Generator Architecture**: 1D Transposed Convolution network trained exclusively on pathological FHR+UC paired windows.
- **Training Signal**: Wasserstein distance with gradient penalty (WGAN-GP, $\lambda = 10$) for training stability.
- **Output**: **1,410 synthetic pathological FHR+UC traces** (3× the original pathological count per fold).
- **Temporal Fidelity**: Unlike SMOTE, TimeGAN preserves physiologically meaningful sequential dynamics in generated traces— including late deceleration morphology and the temporal delay between contraction peaks and FHR nadir.
- **Application Scope**: Augmentation was applied independently per training fold inside the cross-validation loop, ensuring no synthetic samples from validation or test windows contaminated evaluation.

Following TimeGAN augmentation, standard time-series jitter, scaling, and time-warping transformations (2× expansion) were applied to the combined real+synthetic training set for additional regularization.

E. Cross-Validation Protocol

To ensure robust and unbiased evaluation, all model training and evaluation was performed under **Stratified 5-Fold Cross-Validation**:

- **Stratification**: Each fold preserves the ≈7.25% pathological class proportion.
- **Evaluation paradigm**: Out-of-Fold (OOF) evaluation—each sample is predicted exactly once by a model that did not observe it during training, producing an unbiased estimate of generalization performance.
- **Leakage prevention**: All data-dependent transformations (normalization statistics, CSP filter training, TimeGAN training, tabular imputation values) are computed exclusively from the training portion of each fold and applied to the held-out validation portion.
- **No independent held-out test set**: Given the limited dataset size (552 recordings), reserving a separate test partition would substantially reduce training data. The OOF paradigm provides an equivalent statistical guarantee without this sacrifice.
- **Reproducibility**: All stochastic operations were seeded with `random_state=42`.

IV. PROPOSED ARCHITECTURE: ATTENTIONFUSIONRESNET

NeuroFetal AI (v5.0) introduces a tri-modal deep learning architecture, designated as **AttentionFusionResNet**. Unlike conventional CTG classifiers that rely exclusively on the FHR continuous waveform, this architecture combines three distinct clinical representations: (1) Raw 1D Temporal FHR Signals, (2) Static Maternal Clinical Tabular Data, and (3) FHR-UC Spatial Pattern Features (CSP).

The network is structurally composed of three independent feature extraction branches that converge at a textitCross-Modal Attention Fusion (CMAF) module, followed by a probabilistic classification head with Monte Carlo (MC) Dropout for epistemic uncertainty quantification.

A. FHR Temporal Encoder Branch

The primary signal branch processes the 20-minute FHR sequence $\mathbf{X}_{\text{FHR}} \in \mathbb{R}^{1200\times 1}$. The backbone is a 1D Residual Network (ResNet) enhanced with Squeeze-and-Excitation (SE) blocks [23] and Multi-Head Self-Attention.

The encoder begins with a large receptive field convolution (kernel size 7, stride 2) and max pooling to rapidly downsample the uninformative high-frequency noise. This is followed by three stages of cascading residual blocks. Let $\mathbf{x}_l$ denote the input to the l-th residual block. The block operation is defined as:

$$\mathbf{h}_l = \sigma(\text{BN}(\mathbf{W}_{l,2} * \sigma(\text{BN}(\mathbf{W}_{l,1} * \mathbf{x}_l)))) \tag{4}$$

$$\mathbf{x}_{l+1} = \text{ReLU}(\mathbf{x}_l + \text{SE}(\mathbf{h}_l)) \tag{5}$$

where $*$ denotes 1D convolution, BN is Batch Normalization, and $\text{SE}(\cdot)$ represents the channel-wise attention recalibration.

To mitigate overfitting on the small CTU-UHB dataset, we implement **Stochastic Depth** (Spatial Dropout 1D) within the residual paths, randomly dropping entire representation channels during training with a linearly increasing rate ($p \in [0.05, 0.15]$). Following the residual stages, Temporal Multi-Head Attention extracts long-range dependencies (e.g., relating a baseline shift at minute 2 to a late deceleration at minute 18). Finally, Global Average Pooling (GAP) reduces the temporal sequence to a dense vector $\mathbf{v}_{\text{FHR}} \in \mathbb{R}^{192}$.

B. Clinical Context and CSP Branches

Medical decision-making is heavily context-dependent; a moderate deceleration is interpreted differently in a 40-week nulliparous woman compared to a 36-week multiparous woman.

The 16-dimensional tabular vector $\mathbf{X}_{\text{tab}}$ (comprising maternal demographics and extracted signal statistics such as Signal-to-Noise ratio and UC frequency) is processed through a two-layer Multi-Layer Perceptron (MLP) with ReLU activations and heavy dropout ($p = 0.3$) to yield the clinical embedding $\mathbf{v}_{\text{tab}} \in \mathbb{R}^{128}$.

Concurrently, the 19-dimensional Common Spatial Pattern vector $\mathbf{X}_{\text{CSP}}$, which encapsulates the discriminative variance between the FHR and UC contraction cycles, is processed through an identical, parallel MLP structure to yield $\mathbf{v}_{\text{CSP}} \in \mathbb{R}^{128}$.

C. Cross-Modal Attention Fusion (CMAF)

Existing late-fusion architectures often naively concatenate or multiply modal embeddings [4]. We propose a dynamic **Cross-Modal Attention Fusion (CMAF)** layer where the FHR temporal embedding attends to the FHR-UC spatial patterns, explicitly gated by the maternal clinical context.

The CMAF mechanism projects the embeddings into Query ($\mathbf{Q}$), Key ($\mathbf{K}$), and Value ($\mathbf{V}$) spaces:

$$\mathbf{Q} = \mathbf{W}_Q \mathbf{v}_{\text{FHR}} \tag{6}$$

$$\mathbf{K} = \mathbf{W}_K \mathbf{v}_{\text{CSP}} \tag{7}$$

$$\mathbf{V} = \mathbf{W}_V \mathbf{v}_{\text{CSP}} \tag{8}$$

Cross-modal attention scores are computed via scaled dot-product:

$$\mathbf{A} = \text{Softmax}\left(\frac{\mathbf{Q}\mathbf{K}^T}{\sqrt{d_k}}\right)\mathbf{V} \tag{9}$$

The attention output is gated by a learned, sigmoid-activated projection of the clinical tabular data:

$$\mathbf{G}_{\text{clinical}} = \sigma(\mathbf{W}_G \mathbf{v}_{\text{tab}}) \tag{10}$$

$$\mathbf{v}_{\text{fusion}} = \text{LayerNorm}(\mathbf{v}_{\text{FHR}} + \text{Dropout}(\mathbf{A} \odot \mathbf{G}_{\text{clinical}})) \tag{11}$$

This gating mechanism enables the network to actively modulate how much weight it assigns to spatial contraction patterns based on the patient's specific clinical risk factors.

D. Classification and Uncertainty Head

The fused embedding $\mathbf{v}_{\text{fusion}}$ is passed to a classification head comprising two dense layers. To estimate epistemic (model) uncertainty, we apply **Monte Carlo (MC) Dropout** [17].

Unlike standard dropout which is disabled during inference, target dropout layers ($p = 0.3$) remain active. For clinical prediction, the system performs $T = 20$ stochastic forward passes for a single patient trace. The final predictive probability $\hat{p}$ is the mean of these passes, while the predictive variance σ^2 provides a measure of diagnostic uncertainty, flagging cases that require immediate human obstetrician review.

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